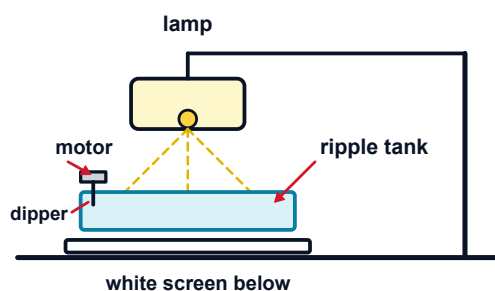


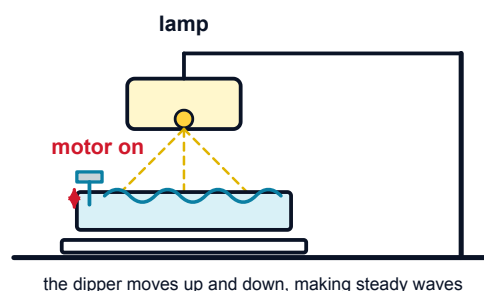
Waves in a ripple tank

A student measures the speed of the waves in a ripple tank.

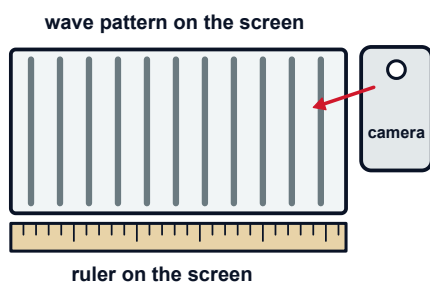
- 1** Set up the ripple tank with a lamp above it and a white screen below.



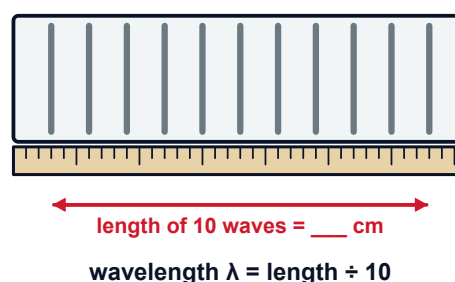
- 2** Switch on the motor so that the dipper makes steady waves.



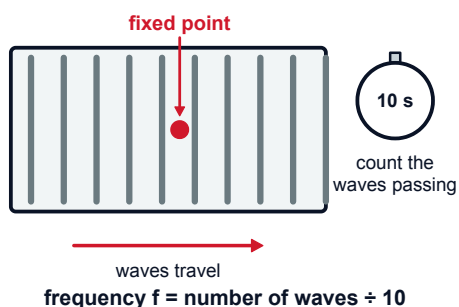
- 3** Place a ruler on the screen and photograph the wave pattern.



- 4** Measure the length of 10 waves on the photograph and divide by 10 to find the wavelength.



- 5** Count the waves passing a point in 10 seconds and divide by 10 to find the frequency.



- 6** Calculate the wave speed: frequency $\times$ wavelength.

$$v = f \times \lambda$$

wave speed = frequency $\times$ wavelength
m/s = Hz $\times$ m

use f from step 5 and λ from step 4

Note: The frequency can be found before or after the wavelength.